1.Write a C program to simulate the following CPU scheduling algorithm to find turnaround time and waiting time.
a) FCFS

CODE:

```
#include <stdio.h>

int main() {
    int n, i, choice;
    int at[10], bt[10], ct[10], tat[10], wt[10];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for(i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
    }

    printf("\nSelect Type:\n");
    printf("1. Non-Preemptive\n");
    printf("2. Preemptive\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    if(choice == 1) {
        // FCFS Scheduling
        ct[0] = at[0] + bt[0];

        for(i = 1; i < n; i++) {
            if(ct[i-1] < at[i])
                ct[i] = at[i] + bt[i];
            else
                ct[i] = ct[i-1] + bt[i];
        }

        for(i = 0; i < n; i++) {
            tat[i] = ct[i] - at[i];
            wt[i] = tat[i] - bt[i];
        }

        printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");
        for(i = 0; i < n; i++) {
            printf("P%d\t%d\t%d\t%d\t%d\t%d\n",
            i+1, at[i], bt[i], ct[i], tat[i], wt[i]);
        }
```

```
    }
    else {
        printf("FCFS does not support Preemptive scheduling\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter number of processes: 2
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 0 4

Select Type:
1. Non-Preemptive
2. Preemptive
Enter choice: 1

P       AT      BT      CT      TAT     WT
P1      0       5       5       5       0
P2      0       4       9       9       5

Process returned 0 (0x0)   execution time : 28.946 s
Press any key to continue.
```

b) SJF (Preemptive and Non-Preemptive)

CODE:

```
#include <stdio.h>
#include <limits.h>

int main() {
    int n,i,time=0,done=0,choice;
    int at[10],bt[10],rt[10],ct[10],wt[10],tat[10];

    printf("Enter number of processes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++){
        printf("AT and BT of P%d: ",i+1);
        scanf("%d%d",&at[i],&bt[i]);
        rt[i]=bt[i];
    }

    printf("\nSelect Type:\n");
    printf("1. Non-Preemptive\n");
    printf("2. Preemptive\n");
    printf("Enter choice: ");
    scanf("%d", &choice);

    if(choice==1){   // Non-preemptive
        int vis[10]={0};
        while(done<n){
            int min=INT_MAX,p=-1;
            for(i=0;i<n;i++)
                if(at[i]<=time && !vis[i] && bt[i]<min){
                    min=bt[i]; p=i;
                }
            if(p==-1){
                time++;
                continue;
            }

            time+=bt[p];
            ct[p]=time;
            vis[p]=1;
            done++;
        }
    }
    else{  // Preemptive (SRTF)
        while(done<n){
            int min=INT_MAX,p=-1;
            for(i=0;i<n;i++)
```

```
            if(at[i]<=time && rt[i]>0 && rt[i]<min){
                min=rt[i]; p=i;
            }
        if(p==-1){
            time++;
            continue;
        }

        rt[p]--;
        time++;

        if(rt[p]==0){
            ct[p]=time;
            done++;
        }
      }
    }

    float tat_total=0;
    float wt_total=0;

    printf("\nP\tAT\tBT\tCT\tTAT\tWT\n");
    for(i=0;i<n;i++){
        tat[i]=ct[i]-at[i];
        tat_total+=tat[i];
        wt[i]=tat[i]-bt[i];
        wt_total+=wt[i];
        printf("P%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],ct[i],tat[i],wt[i]);
    }
    printf("Average tat: %f",(tat_total)/n);
    printf("Average wt: %f",(wt_total)/n);
    return 0;
}
```

OUTPUT:

```
Enter number of processes: 2
AT and BT of P1: 0 5
AT and BT of P2: 0 4

Select Type:
1. Non-Preemptive
2. Preemptive
Enter choice: 1

P       AT      BT      CT      TAT     WT
P1      0       5       9       9       4
P2      0       4       4       4       0

Process returned 0 (0x0)   execution time : 12.355 s
Press any key to continue.
```

```
Enter number of processes: 2
AT and BT of P1: 0 5
AT and BT of P2: 0 4

Select Type:
1. Non-Preemptive
2. Preemptive
Enter choice: 2

P       AT      BT      CT      TAT     WT
P1      0       5       9       9       4
P2      0       4       4       4       0

Process returned 0 (0x0)   execution time : 9.342 s
Press any key to continue.
```

c) Priority (Preemptive and Non-Preemptive)

```c
#include <stdio.h>
#include <stdlib.h>

int main(){
    int n,i,choice,count=0,time=0;
    printf("Enter no. of processes: \n");
    scanf("%d",&n);

    int at[n],bt[n],pr[n],ct[n],tat[n],wt[n],rt[n];
    for(i=0;i<n;i++){
        printf("Enter AT,BT and Priority for P%d: ",i+1);
        scanf("%d %d %d",&at[i],&bt[i],&pr[i]);
        rt[i]=bt[i];
    }
    printf("Select Type:\n");
    printf("1. Non preemptive\n");
    printf("2. Preemptive\n");
    printf("Enter choice: ");
    scanf("%d",&choice);

    if(choice==1){
        int completed[n];
        for(i=0;i<n;i++)
            completed[i]=0;
```

```
    while(count<n){
        int idx=-1;
        int highest=9999;

        for(i=0;i<n;i++){
            if(at[i]<=time && completed[i]==0){
                if(pr[i]<highest){
                    highest=pr[i];
                    idx=i;
                }
            }
        }

        if(idx!=-1){
            time+=bt[idx];

            ct[idx]=time;
            tat[idx]=ct[idx]-at[idx];
            wt[idx]=tat[idx]-bt[idx];

            completed[idx]=1;
            count++;
        }
        else time++;
    }
}

else if(choice==2){
    while(count<n){
        int idx=-1;
        int highest=9999;

        for(i=0;i<n;i++){
            if(at[i]<=time && rt[i]>0){
                if(pr[i]<highest){
                    highest=pr[i];
                    idx=i;
                }
            }
        }

        if(idx!=-1){
            rt[idx]--;
            time++;

            if(rt[idx]==0){
                ct[idx]=time;
                tat[idx]=ct[idx]-at[idx];
```

```
                wt[idx]=tat[idx]-bt[idx];
                count++;
            }
        }
        else time++;
    }
}

else printf("Enter from the choices given!\n");

printf("\nP\tAT\tBT\tPR\tCT\tTAT\tWT\n");

float tatsum=0,wtsum=0;

for(i=0;i<n;i++){
    tatsum+=tat[i];
    wtsum+=wt[i];
    printf("P%d\t%d\t%d\t%d\t%d\t%d\t%d\n",i+1,at[i],bt[i],pr[i],ct[i],tat[i],wt[i]);
}

printf("\nAvg TAT: %f\nAvg WT: %f\n",tatsum/n,wtsum/n);

return 0;
}
```

OUTPUT:

```
Enter no. of processes:
4
Enter AT,BT and Priority for P1: 0 4 5
Enter AT,BT and Priority for P2: 3 2 6
Enter AT,BT and Priority for P3: 5 5 1
Enter AT,BT and Priority for P4: 2 3 3
Select Type:
1. Non preemptive
2. Preemptive
Enter choice: 1

P       AT      BT      PR      CT      TAT     WT
P1      0       4       5       4       4       0
P2      3       2       6       14      11      9
P3      5       5       1       12      7       2
P4      2       3       3       7       5       2

Avg TAT: 6.750000
Avg WT: 3.250000

Process returned 0 (0x0)   execution time : 28.057 s
Press any key to continue.
```

```
Enter no. of processes:
4
Enter AT,BT and Priority for P1: 0 4 5
Enter AT,BT and Priority for P2: 3 2 6
Enter AT,BT and Priority for P3: 5 5 1
Enter AT,BT and Priority for P4: 2 3 3
Select Type:
1. Non preemptive
2. Preemptive
Enter choice: 2

P       AT      BT      PR      CT      TAT     WT
P1      0       4       5       12      12      8
P2      3       2       6       14      11      9
P3      5       5       1       10      5       0
P4      2       3       3       5       3       0

Avg TAT: 7.750000
Avg WT: 4.250000

Process returned 0 (0x0)   execution time : 26.453 s
Press any key to continue.
```